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FIT3155: Advanced Algorithms and Data Structures

Week 2: **Linear-time string pattern matching (Boyer-Moore, Knuth-Morris-Pratt algorithms)**

Faculty of Information Technology, Monash University

Last week

We looked at...

The Z-algorithm

The Z-algorithm is a general (preprocessing) algorithm to Computes Z_i-values of a string **str** in $\mathcal{O}(|\mathbf{str}|)$-time (refer week 1's lecture).

The exact pattern matching problem, addressed using Z-algorithm

Given a reference text **txt**$[1 \ldots n]$ and a pattern **pat**$[1 \ldots m]$, we used the Z-algorithm to find **ALL** occurrences, if any, of **pat** in **txt** in $\mathcal{O}(m + n)$ time.

Week 1 recap: Naïve approach

Example

	1	2	3	4	5	6	7	8	9	10	11	12	13
txt:	x	a	b	x	y	a	b	x	y	a	b	x	z
pat:	a	b	x	y	a	b	x	z					
iteration-1	✗												
pat:		a	b	x	y	a	b	x	z				
iteration-2		✓	✓	✓	✓	✓	✓	✓	✗				
pat:			a	b	x	y	a	b	x	z			
iteration-3			✗										
pat:				a	b	x	y	a	b	x	z		
iteration-4				✗									
pat:					a	b	x	y	a	b	x	z	
iteration-5					✗								
pat:						a	b	x	y	a	b	x	z
iteration-6						✓	✓	✓	✓	✓	✓	✓	✓

Overall 20 comparisons in the naïve approach, on this example.

Week 1 recap: Main ideas to improve naïve pattern matching

We imagined ideas to:

Shift more purposefully: Shift **pat** by > 1 position after each iteration, whenever possible.

Shift always safely: Never shift so far as to miss any exact occurrence of **pat** in **txt**.

Week 1 recap: Main ideas to improve naïve pattern matching

We imagined ideas to:

Shift more purposefully: Shift **pat** by > 1 position after each iteration, whenever possible.

Shift always safely: Never shift so far as to miss any exact occurrence of **pat** in **txt**.

Two specific algorithms (considered classics in computer science) implement these ideas efficiently and in concrete ways:

1. Boyer-Moore
2. Knuth-Morris-Pratt

The Boyer-Moore Algorithm

Comments:

- Defines the standard benchmark for string pattern matching.
- Many find/search utilities that come packaged within programming languages/operating systems rely on this algorithm.
- Incorporated within GNU's implementation of **`grep`**.

Boyer-Moore Algorithm – Introduction

Boyer-Moore algorithm incorporate three main ideas:

1. right-to-left scanning
2. bad-character shift rule
3. good-suffix shift rule
   - matched-prefix shift rule (a variant of the good-suffix rule used in special scenarios)

We will discuss each of these in detail!

Boyer-Moore: General structure of the algorithm

- **`Preprocessing phase`** (performed once at the start of the algorithm solely using **`pat`**):
  - Preprocess **`pat`** to compute three data structures to support
    1. shifting under bad-character shift rule.
    2. shifting under good-suffix shift rule.
    3. shifting under matched-prefix shift rule.

       matched-prefix shifting is used in any iteration when (i) either, another instance of good suffix is not found in the **`pat`**, (ii) or after **`pat`** matches the entire region in the **`txt`**.
- **`Iterative search phase`** (i.e. iterated until end of **`txt`**):
  - In each iteration where **`pat`** is under some region of **`txt`**, compare/scan **`pat`** and **`txt`** characters of that region from right to left.
  - Shift **`pat`** rightwards safely* under **`txt`** by the maximum number of positions allowed by:
    - ★ either bad-character shift rule
    - ★ or good-suffix/matched-prefix shift rule.

*A shift is safe if and only if such a shift does not skip past (and miss) any exact occurrence of **`pat`** in **`txt`**.

Right-to-left scanning

For any comparison of **pat**$[1 \ldots m]$ against **txt**$[j \ldots j+m-1]$, the Boyer-Moore algorithm checks/scans for matched characters right to left.

Example: right to left scanning (in some arbitrary iteration)

```
     1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
txt: y p b c t b y a b p  q  y  c  t  b  p  q  p
                   ←
         1 2 3 4 5 6 7
pat:     t b a p y a b
```

Scanning from right to left in the above example:

- Compare **pat**$[7] \equiv$ b with **txt**$[9] \equiv$ b: match.
- Compare **pat**$[6] \equiv$ a with **txt**$[8] \equiv$ a: match.
- Compare **pat**$[5] \equiv$ y with **txt**$[7] \equiv$ y: match.
- Compare **pat**$[4] \equiv$ p with **txt**$[6] \equiv$ b: mismatch.

After encountering a mismatch in any right-to-left scanning, Boyer-Moore uses information from the character pairs already compared in the current iteration to determine the maximum **safe** rightward shift (by ≥ 1 **positions**) of the pattern **pat** under the text **txt**. It employs two shift rules to do this: (1) the bad-character shift rule and (2) the good-suffix shift rule.

Preprocessing 'Bad-character' shift rule data structure

- Let **pat**$[1 \ldots m]$ and **txt**$[1 \ldots n]$ be strings from the alphabet Σ.
- Preprocess **pat**, and store for each character $x \in \Sigma$, the **rightmost** position of occurrence of character x in **pat**.
  - Call this position, $R(x)$.
  - When x does not occur in **pat**, $R(x) = 0$ (in 1-based indexing of **pat**).

$R(x)$ values stored in an array data structure

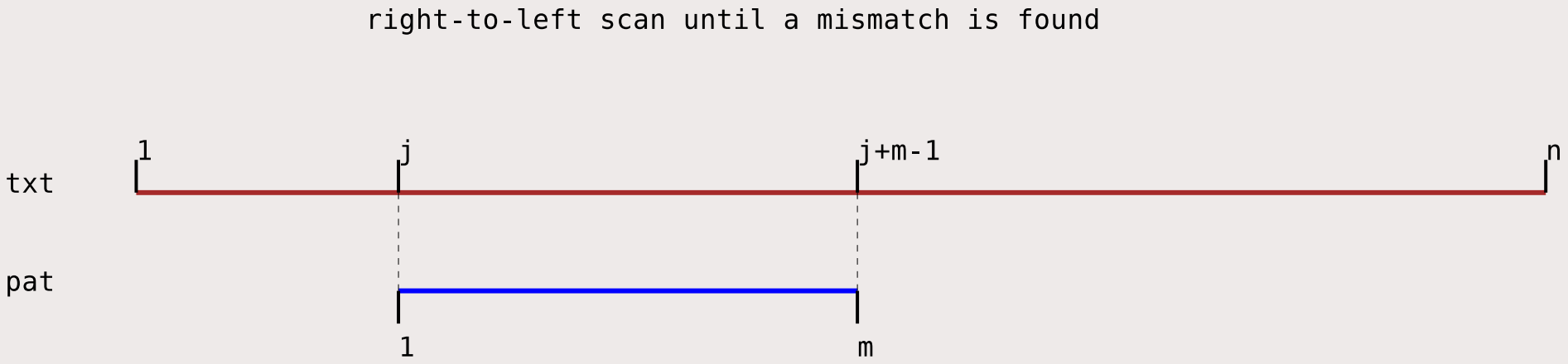
The $R(x)$ values for **pat**$[1 \ldots 7] =$ $\overset{1}{\mathtt{t}}\ \overset{2}{\mathtt{b}}\ \overset{3}{\mathtt{a}}\ \overset{4}{\mathtt{p}}\ \overset{5}{\mathtt{y}}\ \overset{6}{\mathtt{a}}\ \overset{7}{\mathtt{b}}$ are:

$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	y	...
$R(x)$	6	7	0	...	4	0	...	1	...	5	...

Complexity

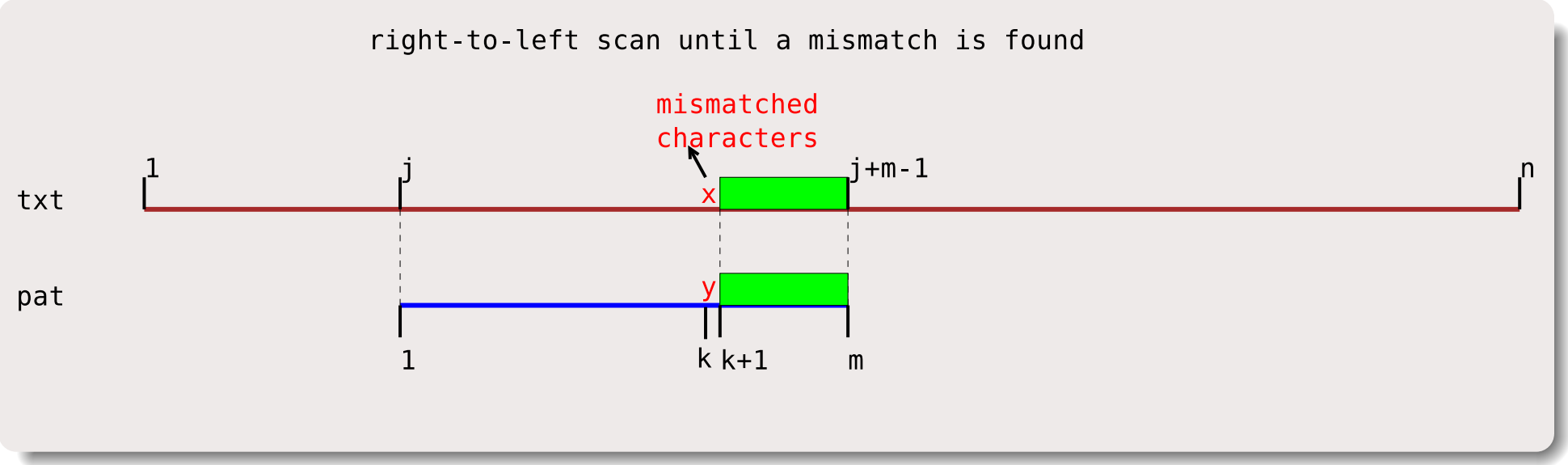
Precomputing $R(x)$ values for any given **pat** requires $\mathcal{O}(|\Sigma|)$ space and $\mathcal{O}(m + |\Sigma|)$ time. After computing this array, during any iteration of Boyer-Moore's **txt** searching phase, it takes $\mathcal{O}(1)$-time to lookup this array and get the rightmost position of any possible (bad-)character in **txt** and derive how many positions to shift **pat** by rightwards to start the next iteration.

Formalizing 'Bad character' shift rule



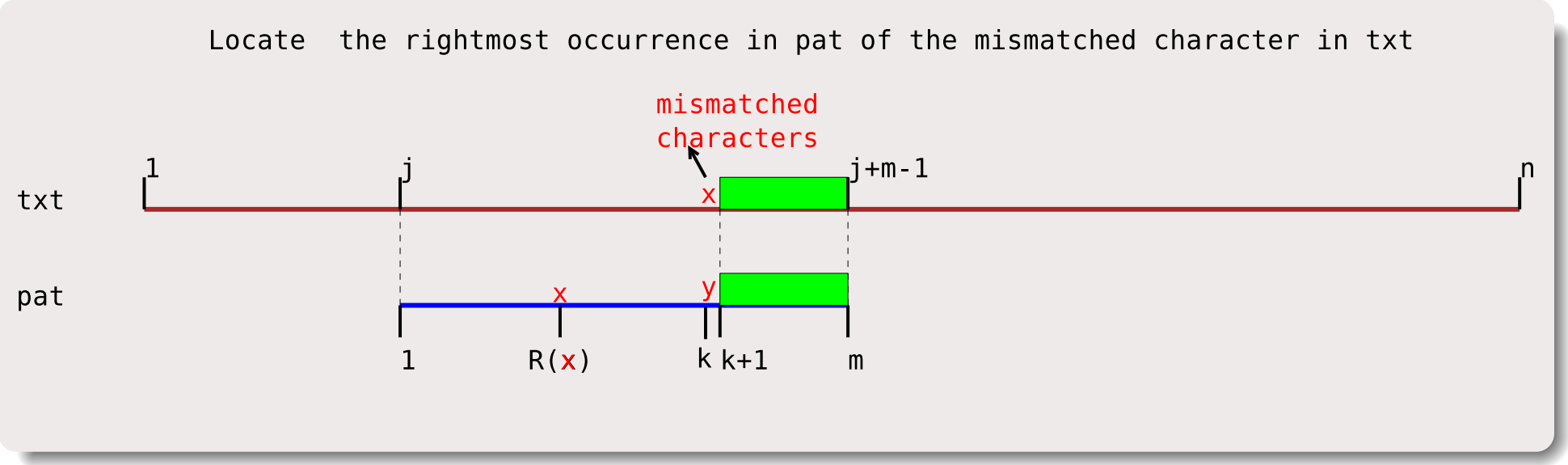
- More abstractly, in some iteration, if **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via the right-to-left scan, and

Formalizing 'Bad character' shift rule



- More abstractly, in some iteration, if **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via the right-to-left scan, and
- ...the kth character of **txt** counting from position j, i.e. $x =$ **txt**$[j+k-1]$, be the bad character that mismatched with the kth character of the pattern **y** $=$ **pat**$[k]$, then

Formalizing 'Bad character' shift rule



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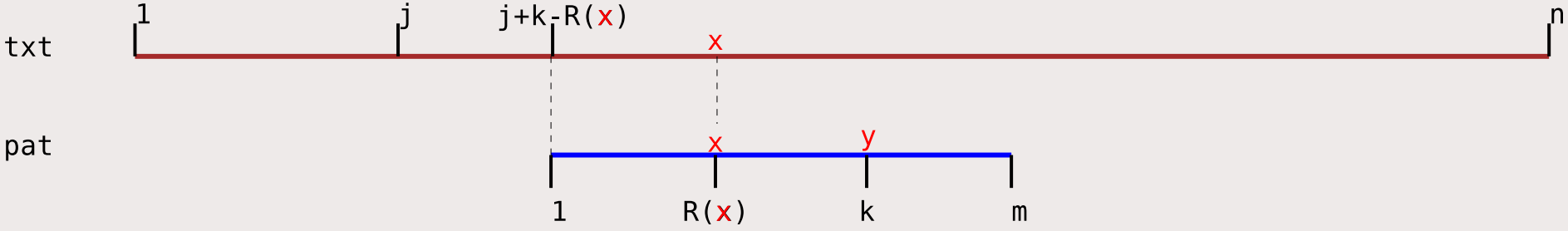
- ...the kth character of **txt** counting from position j, i.e. $x =$ **txt**$[j+k-1]$, be the bad character that mismatched with the kth character of the pattern **y** $=$ **pat**$[k]$, then
- ...using the preprocessed $R(x)$ data structure, (potentially) locate the rightmost occurrence $R(x)$ in **pat** of the bad character x.

Formalizing 'Bad character' shift rule

```
Shift pat so that the rightmost occurrence in pat of the mismatched character in txt
will aligned after the shift
```



- More abstractly, in some iteration, if **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via the right-to-left scan, and
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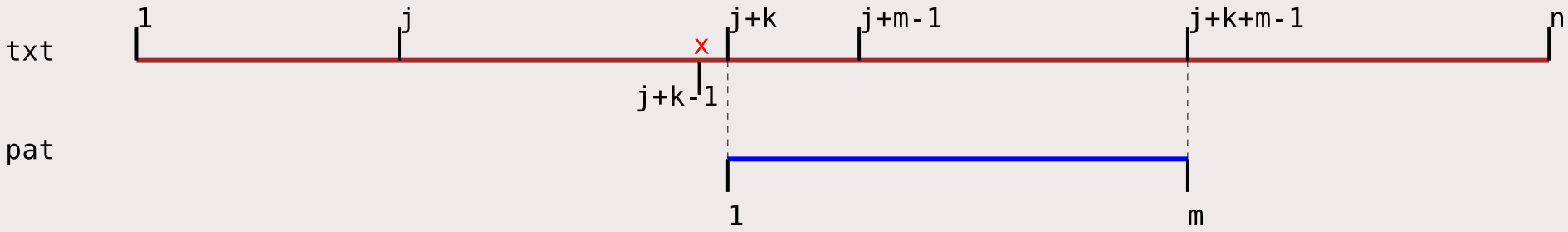
- ...using the preprocessed $R(x)$ data structure, (potentially) locate the rightmost occurrence $R(x)$ in **pat** of the bad character x.

Bad-character shift RULE: If $R(x) < k$, shift **pat** rightwards by $k - R(x)$ positions and start the next iteration. Else, one has no choice with this basic array data structure but to shift **pat** rightwards naïvely by 1 position.

Formalizing 'Bad character' shift rule

```
Assume the bad character (x) in text were NOT found in pat. This implies R(x) = 0.
The rule automatically implies that the shift of pat goes past the mismatched
position in txt.
```



- More abstractly, in some iteration, if **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via the right-to-left scan, and
- ...the kth character of **txt** counting from position j, i.e. $x =$ **txt**$[j+k-1]$, be the bad character that mismatched with the kth character of the pattern **y** $=$ **pat**$[k]$, then
- ...using the preprocessed $R(x)$ data structure, (potentially) locate the rightmost occurrence $R(x)$ in **pat** of the bad character x.

Bad-character shift RULE: If $R(x) < k$, shift **pat** rightwards by $k - R(x)$ positions and start the next iteration. Else, one has no choice with this basic array data structure but to shift **pat** rightwards naïvely by 1 position.

- Note: The above rule automatically implies that if x does not occur in **pat**$[1..m]$ (i.e., $R(x) = 0$), then the **pat** gets shifted one position past the point of mismatch in **txt**.

Bad character shift rule example

Consider the following pattter and text strings:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
	1	2	3	4	5	6	7											
pat:	t	b	a	p	y	a	b											

Note: **pat** string slightly different from previous.

The $R(x)$ values for the pattern are:

$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	y	...
$R(x)$	6	7	0	...	4	0	...	1	...	5	...

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
iteration 1							✗											
pat:	t	b	a	p	y	a	b											
	1	2	3	4	5	6	7											

Shift by $\max(1, k - R(x)) = \max(1, 7 - 5) = 2$ positions to start next iteration.

Bad character shift rule example

Consider the following pattter and text strings:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
	1	2	3	4	5	6	7											
pat:	t	b	a	p	y	a	b											

Note: **pat** string slightly different from previous.

The $R(x)$ values for the pattern are:

$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	y	...
$R(x)$	6	7	0	...	4	0	...	1	...	5	...

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
iteration 2						✗	✓	✓	✓									
pat:			t	b	a	p	y	a	b									
			1	2	3	4	5	6	7									

Shift by $\max(1, k - R(x)) = \max(1, 4 - 7) = 1$ position to start next iteration.

Bad character shift rule example

Consider the following pattter and text strings:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
	1	2	3	4	5	6	7											
pat:	t	b	a	p	y	a	b											

Note: **pat** string slightly different from previous.

The $R(x)$ values for the pattern are:

$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	y	...
$R(x)$	6	7	0	...	4	0	...	1	...	5	...

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
iteration 3										✗								
pat:				t	b	a	p	y	a	b								
				1	2	3	4	5	6	7								

Shift by $\max(1, k - R(x)) = \max(1, 7 - 6) = 1$ position to start next iteration.

Bad character shift rule example

Consider the following pattter and text strings:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
	1	2	3	4	5	6	7											
pat:	t	b	a	p	y	a	b											

Note: **pat** string slightly different from previous.

The $R(x)$ values for the pattern are:

$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	y	...
$R(x)$	6	7	0	...	4	0	...	1	...	5	...

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
iteration 4											✗							
pat:					t	b	a	p	y	a	b							
					1	2	3	4	5	6	7							

Shift by $\max(1, k - R(x)) = \max(1, 7 - 0) = 7$ positions to start next iteration.

Bad character shift rule example

Consider the following pattter and text strings:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
	1	2	3	4	5	6	7											
pat:	t	b	a	p	y	a	b											

Note: **pat** string slightly different from previous.

The $R(x)$ values for the pattern are:

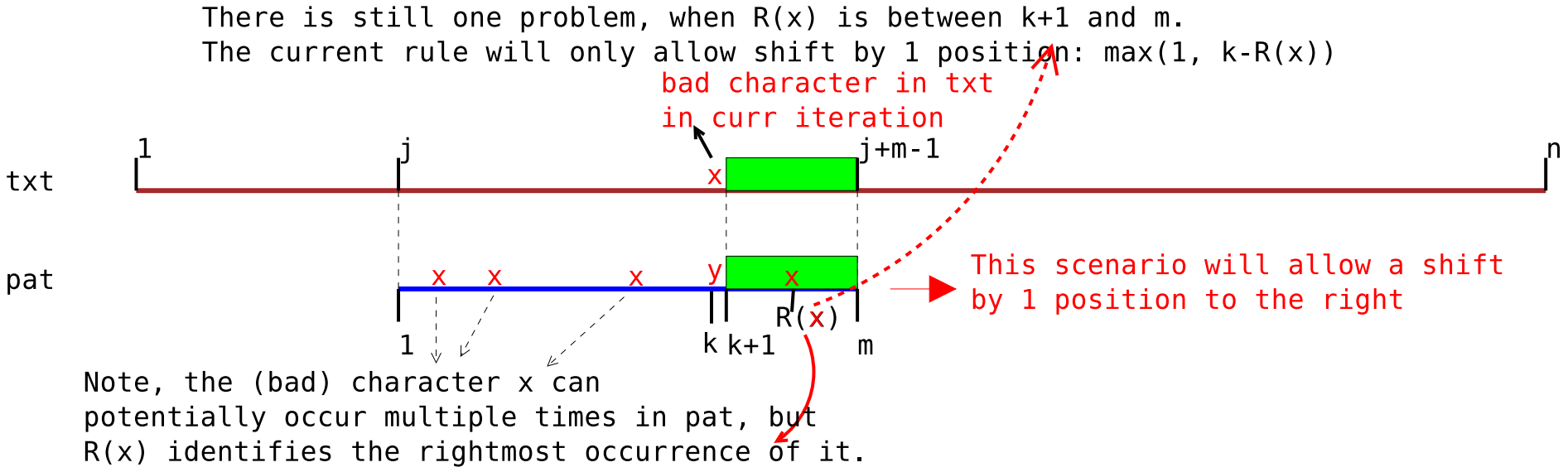
$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	y	...
$R(x)$	6	7	0	...	4	0	...	1	...	5	...

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
txt:	y	p	b	c	t	b	y	a	b	a	c	b	y	t	b	p	q	p
iteration 5																		✗
pat:												t	b	a	p	y	a	b
												1	2	3	4	5	6	7

Shift by $\max(1, k - R(x)) = \max(1, 7 - 4) = 3$ positions to start next iteration (if more text still available; otherwise stop)

Improving further: The extended bad-character shift rule

Recall from the bad character shift rule (see slide 11, last overlay), that when $R(x) > k$, then we shift naïvely by 1 position.

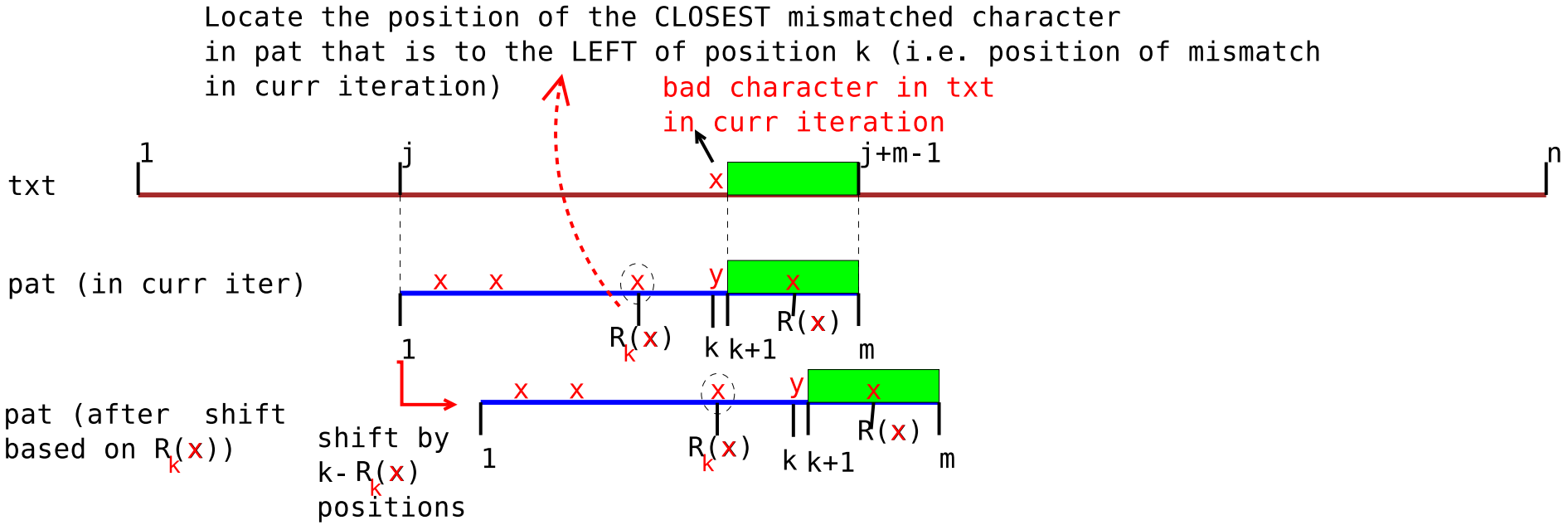


How can we extend the $R(\cdot)$ data structure so that we can do better than that, when any $R(x) > k$?

Improving further: The extended bad-character shift rule

Extended Bad-Character Rule

When a mismatch occurs at some position k in **pat**$[1 \ldots m]$, and the corresponding mismatched character is $x =$ **txt**$[j + k - 1]$, then **shift** **pat**$[1..m]$ to the right so that **the closest x in pat that is to the left of position** k is now below the (previously mismatched) x in **txt**.



Improving further: The extended bad-character shift rule

Preprocessing

- To achieve this, preprocess **pat**$[1 \ldots m]$ to compute $R_k(x)$ values...
  - for each position $1 \leq k \leq m$ in **pat**,
  - and for each character x in the alphabet Σ,

  ...storing the **rightmost** position of x **to the left of** $(<)$ k

In Boyer-Moore's search phase...

- For any possible combinations of k and x, if we denote such a position as $R_k(x)$, then this data strcuture implementing the extended bad character shift rule will ask you to shift **pat safely** by $k - R_k(x)$ positions, when a mismatch occurs at any position k in **pat** with x as the bad-character in the **txt**.

Complexity

The most basic implementation of the $R_k(x)$ data structure is a 2D array/table, taking $\mathcal{O}(m \times |\Sigma|)$ space and time to precompute (example in next slide). In turn, this will provide $\mathcal{O}(1)$ lookup during the search phase.

(Quiz: Can this is implemented in $\mathcal{O}(m+|\Sigma|)$ space? Does this trade-off lookup time?)

Improving further: The extended bad-character shift rule

Self-study: Example of a basic extended $R_k(\cdot)$ data structure

For **pat**$[1 \ldots 7] =$ $\overset{1}{\mathtt{t}}\ \overset{2}{\mathtt{b}}\ \overset{3}{\mathtt{a}}\ \overset{4}{\mathtt{p}}\ \overset{5}{\mathtt{x}}\ \overset{6}{\mathtt{a}}\ \overset{7}{\mathtt{b}}$, one could store the corresponding $R_k(x)$ values in the following rudimentary 2D matrix of size $m \times |\Sigma|$:

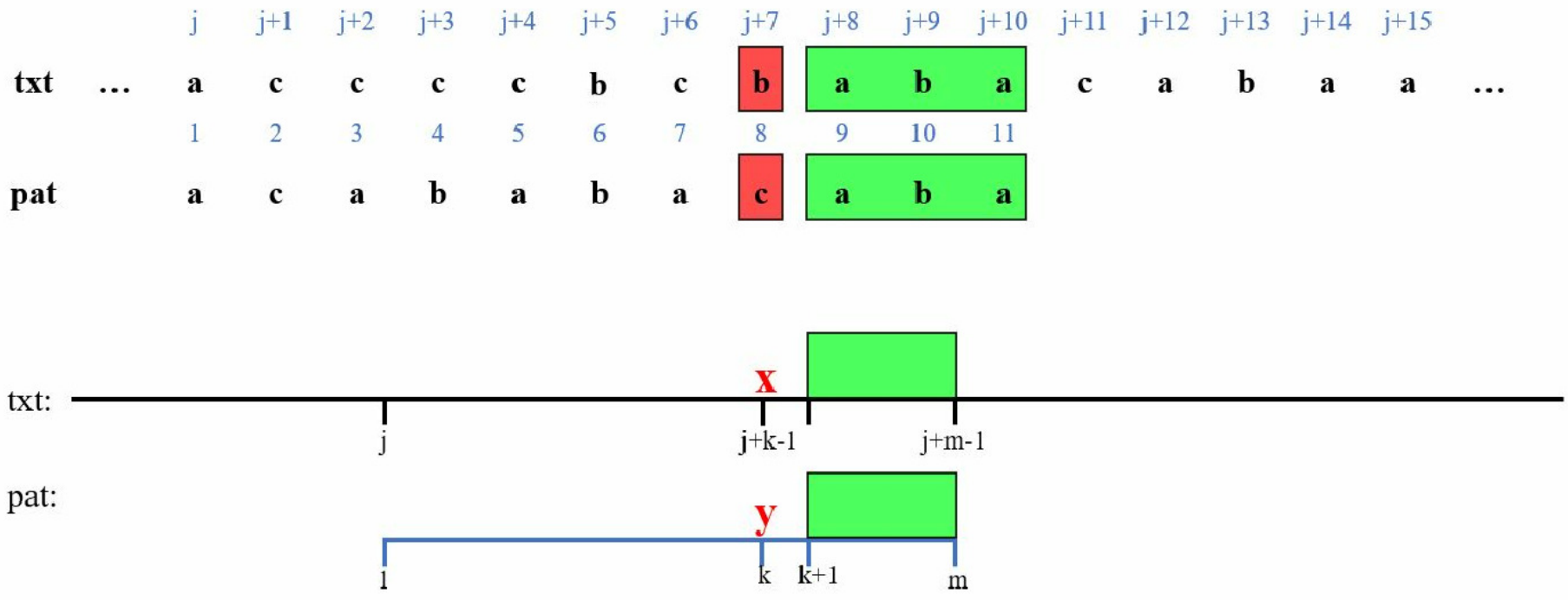
$\forall x \in \Sigma$	a	b	c	...	p	q	...	t	...	x	...
$R_1(x)$	0	0	0	...	0	0	...	0	...	0	...
$R_2(x)$	0	0	0	...	0	0	...	1	...	0	...
$R_3(x)$	0	2	0	...	0	0	...	1	...	0	...
$R_4(x)$	3	2	0	...	0	0	...	1	...	0	...
$R_5(x)$	3	2	0	...	4	0	...	1	...	0	...
$R_6(x)$	3	2	0	...	4	0	...	1	...	5	...
$R_7(x)$	6	2	0	...	4	0	...	1	...	5	...

Boyer-Moore Algorithm – Good suffix rule: Motivation

		j	j+1	j+2	j+3	j+4	j+5	j+6	j+7	j+8	j+9	j+10	j+11	j+12	j+13	j+14	j+15	
txt	**...**	**a**	**c**	**c**	**c**	**c**	**b**	**c**	**b**	**a**	**b**	**a**	**c**	**a**	**b**	**a**	**a**	**...**
		1	2	3	4	5	6	7	8	9	10	11						
pat		**a**	**c**	**a**	**b**	**a**	**b**	**a**	**c**	**a**	**b**	**a**						

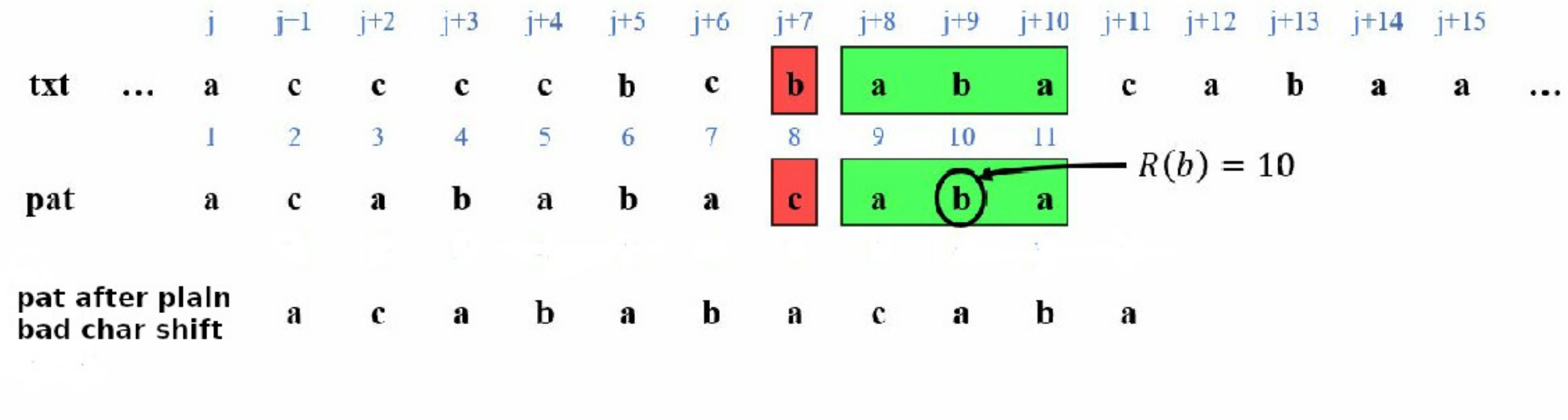
- In some iteration, say **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via right-to-left scan.

Boyer-Moore Algorithm – Good suffix rule: Motivation



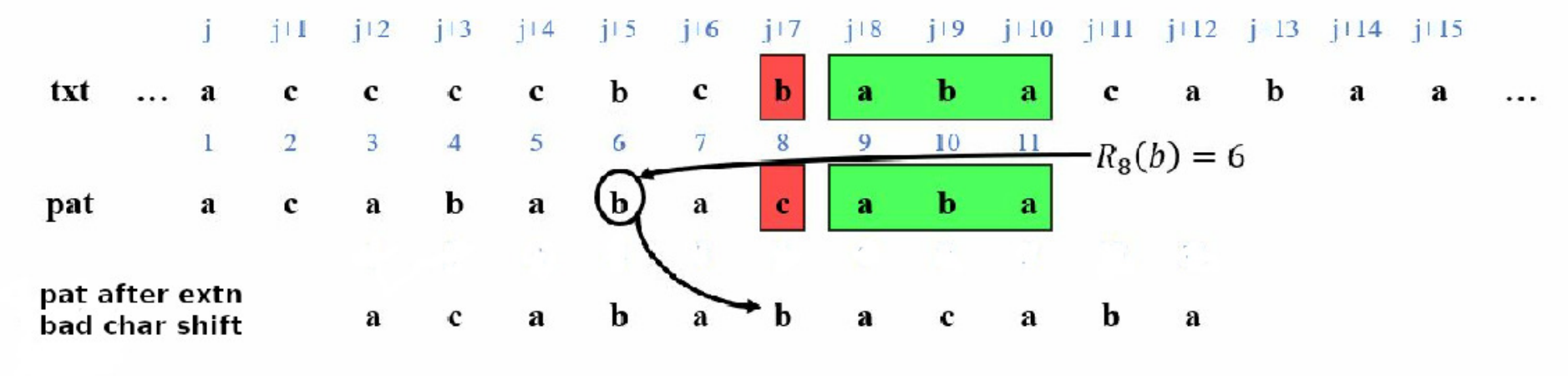
- In some iteration, say **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via right-to-left scan.
- Let the kth character of **txt** counting from j, i.e. $x =$ **txt**$[j+k-1]$, be mismatched with the kth character of the pattern **y** $=$ **pat**$[k]$.

Boyer-Moore Algorithm – Good suffix rule: Motivation



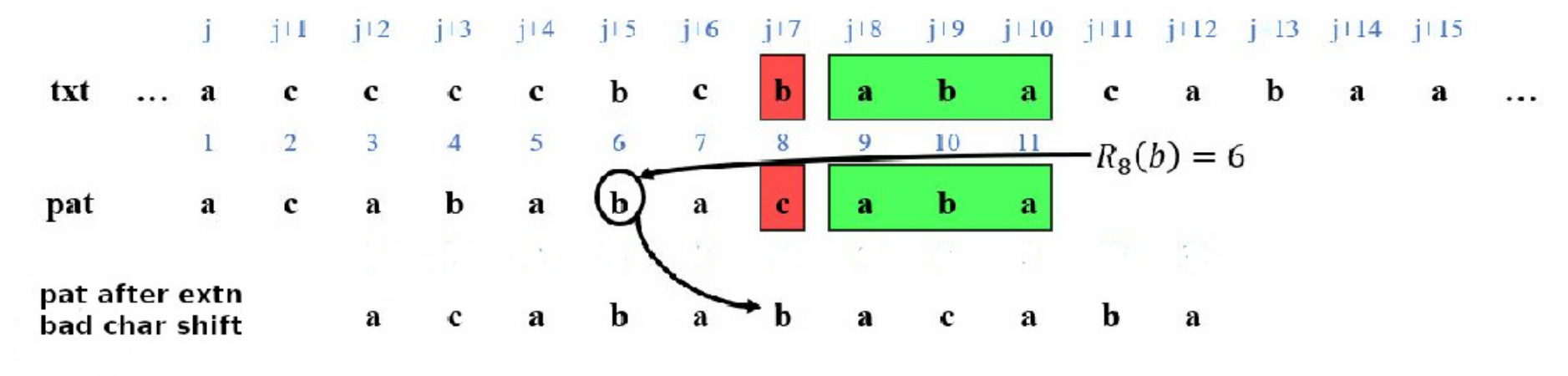
- In some iteration, say $\texttt{txt}[j \ldots j+m-1]$ and $\texttt{pat}[1 \ldots m]$ are being compared via right-to-left scan.
- Let the kth character of **txt** counting from j, i.e. $x = \texttt{txt}[j+k-1]$, be mismatched with the kth character of the pattern $\texttt{y} = \texttt{pat}[k]$.
- The plain $R(\cdot)$ based bad character rule will only suggest a naïve shift of **pat** by 1 position to the right.

Boyer-Moore Algorithm – Good suffix rule: Motivation



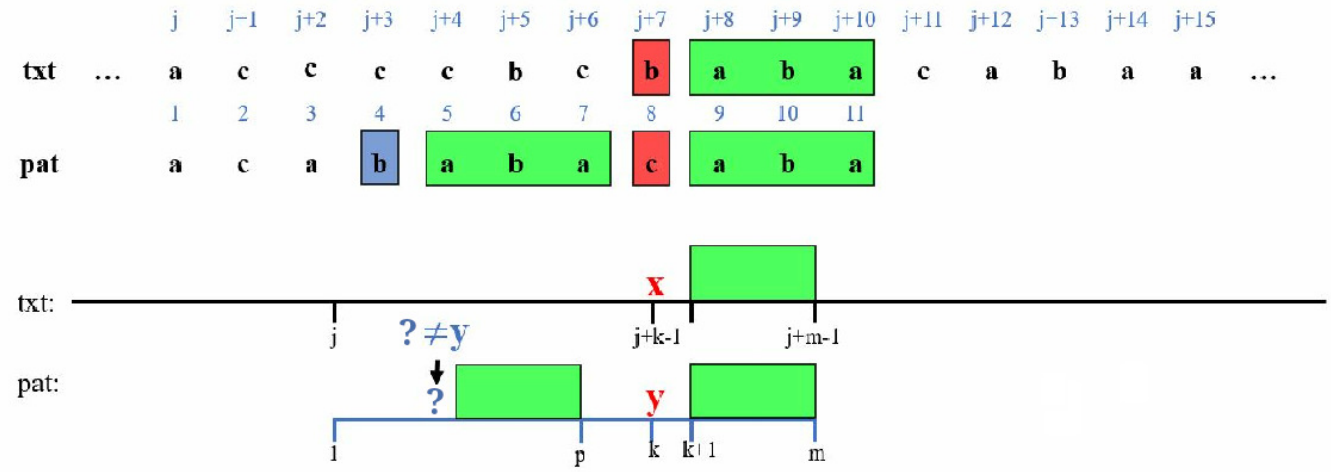
- In some iteration, say **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via right-to-left scan.
- Let the kth character of **txt** counting from j, i.e. $x =$ **txt**$[j+k-1]$, be mismatched with the kth character of the pattern **y** $=$ **pat**$[k]$.
- The plain $R(\cdot)$ based bad character rule will only suggest a naïve shift of **pat** by 1 position to the right.
- The extended $R_k(\cdot)$ based bad character rule suggests a shift of 2 positions to the right.

Boyer-Moore Algorithm – Good suffix rule: Motivation



- In some iteration, say **txt**$[j \ldots j+m-1]$ and **pat**$[1 \ldots m]$ are being compared via right-to-left scan.
- Let the kth character of **txt** counting from j, i.e. $x =$ **txt**$[j+k-1]$, be mismatched with the kth character of the pattern **y** $=$ **pat**$[k]$.
- The plain $R(\cdot)$ based bad character rule will only suggest a naïve shift of **pat** by 1 position to the right.
- The extended $R_k(\cdot)$ based bad character rule suggests a shift of 2 positions to the right.
- The motivation behind Good suffix rule is to explore whether the pattern can **safely** be shifted by more positions than what the extended bad character rule suggests.

Boyer-Moore Algorithm – Good suffix rule: Intuition



- Suppose there exists a position $p < m$ in **pat** such that it is the **endpoint** of the rightmost occurrence of the **substring** such that:
  - $\texttt{pat}[p - m + k + 1 \ldots p] \equiv \texttt{pat}[k + 1 \ldots m]$, and
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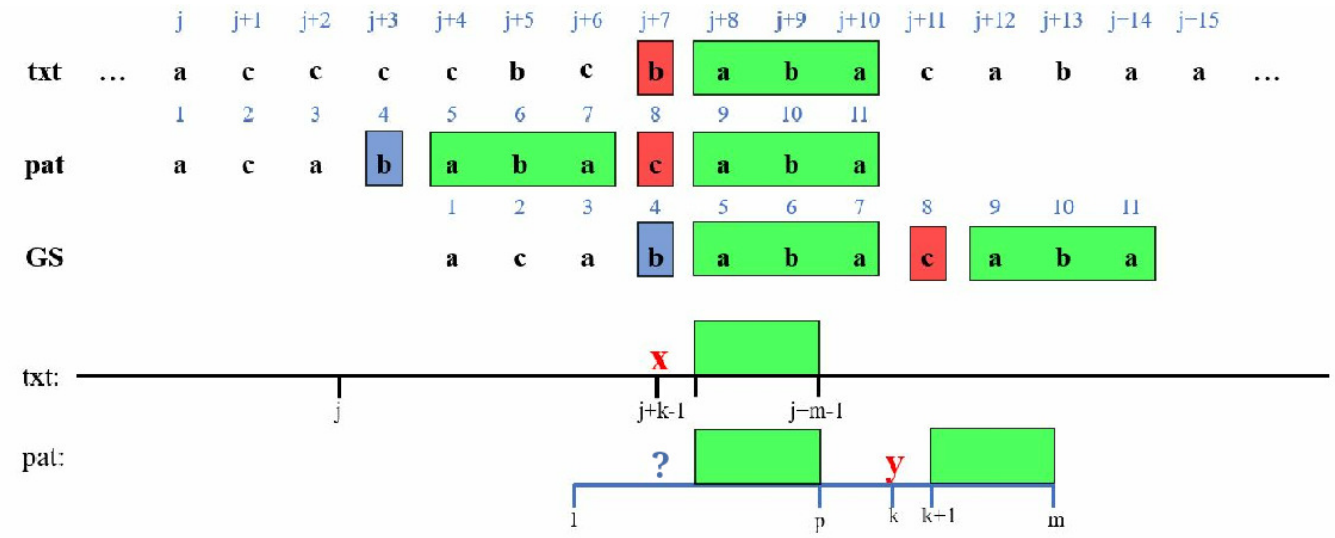

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  - $\texttt{pat}[p - m + k + 1 \ldots p] \equiv \texttt{pat}[k + 1 \ldots m]$, and
  - $\texttt{pat}[p - m + k] \neq \texttt{pat}[k]$,
- then **pat** can be safely and efficiently shifted right under the **txt** by $m - p$ positions, to begin a new iteration.

Boyer-Moore Algorithm – Ideas to implement the 'good suffix' rule efficiently

To efficiently implement this 'good suffix' rule, we take 'inspiration' from the computation of Z_i values in Gusfield's algorithm, and define Z_i^{suffix} (specifically on **pat**) as follows:

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Definition of Z_i^{suffix}

Given a **pat**$[1 \ldots m]$, define Z_i^{suffix} (for each position $1 \leq i < m$) as the **length** of the **longest substring ending at position** i of **pat** that matches its suffix (i.e., **pat**[i-Z_i^{suffix}+1 ... i] = **pat**[m-Z_i^{suffix}+1 ... m]).

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- Note, computation of Z_i^{suffix} values on **pat** can be derived after running Z-algorithm on reverse(**pat**).
- Thus, Z_i^{suffix} values can be computed in $\mathcal{O}(m)$ time for **pat**$[1 \ldots m]$.

Boyer-Moore Algorithm – Implementing 'good suffix' rule efficiently

In fact, for each **suffix** starting at position $j \in [1 \ldots m]$ in **pat**, we want to store the rightmost position p in **pat** such that:

- $\mathtt{pat}[j..m] \equiv \mathtt{pat}[p - Z_p^{\text{suffix}} + 1 \ldots p]$.
- $\mathtt{pat}[j-1] \neq \mathtt{pat}[p - Z_p^{\text{suffix}}]$.

Boyer-Moore Algorithm – Implementing 'good suffix' rule efficiently

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- $\mathbf{pat}[j-1] \neq \mathbf{pat}[p - Z_p^{\text{suffix}}]$.

Store these rightmost positions as $\mathbf{goodsuffix}(j) = p$.

After computing the Z_p^{suffix} values, $\mathbf{goodsuffix}(j)$ values can be computed from them using this simple logic – reason why during self-study.

```
Z^{suffix} := computeZSuffixValues(pat)
m := |pat|
for j from 1 to m+1  do
    goodsuffix(j) := 0
endfor

for p from 1 to m-1  do
    j := m - Z^{suffix}(p) + 1
    goodsuffix(j) := p
endfor
```

Self-study: Example of good suffix values computation

For **pat**$[1 \ldots 11] =$ $\overset{1}{\mathtt{a}}\ \overset{2}{\mathtt{c}}\ \overset{3}{\mathtt{a}}\ \overset{4}{\mathtt{b}}\ \overset{5}{\mathtt{a}}\ \overset{6}{\mathtt{b}}\ \overset{7}{\mathtt{a}}\ \overset{8}{\mathtt{c}}\ \overset{9}{\mathtt{a}}\ \overset{10}{\mathtt{b}}\ \overset{11}{\mathtt{a}}$

Z^{suffix} and goodsuffix values after lines 1-5 of the pseudocode

j	1	2	3	4	5	6	7	8	9	10	11	12
pat	a	c	a	b	a	b	a	c	a	b	a	
Z^{suffix}	1	0	1	0	5	0	3	0	1	0	-	-
goodsuffix	0	0	0	0	0	0	0	0	0	0	0	0

```
1  Z^{suffix} := computeZSuffixValues(pat)
2  m := |pat|
3  for j from 1 to m+1  do
4      goodsuffix(j) := 0
5  endfor
6
7  for p from 1 to m-1  do
8      j := m - Z^{suffix}(p) + 1
9      goodsuffix(j) := p
10 endfor
```

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Z^{suffix} and goodsuffix values after lines 1-10 of the pseudocode

j	1	2	3	4	5	6	7	8	9	10	11	12
pat	a	c	a	b	a	b	a	c	a	b	a	
Z^{suffix}	1	0	1	0	5	0	3	0	1	0	-	-
goodsuffix	0	0	0	0	0	0	5	0	7	0	9	10

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Boyer-Moore Algorithm – Using 'good suffix' rule during search

In any iteration, to use the 'good suffix' rule, the following cases have to handled:

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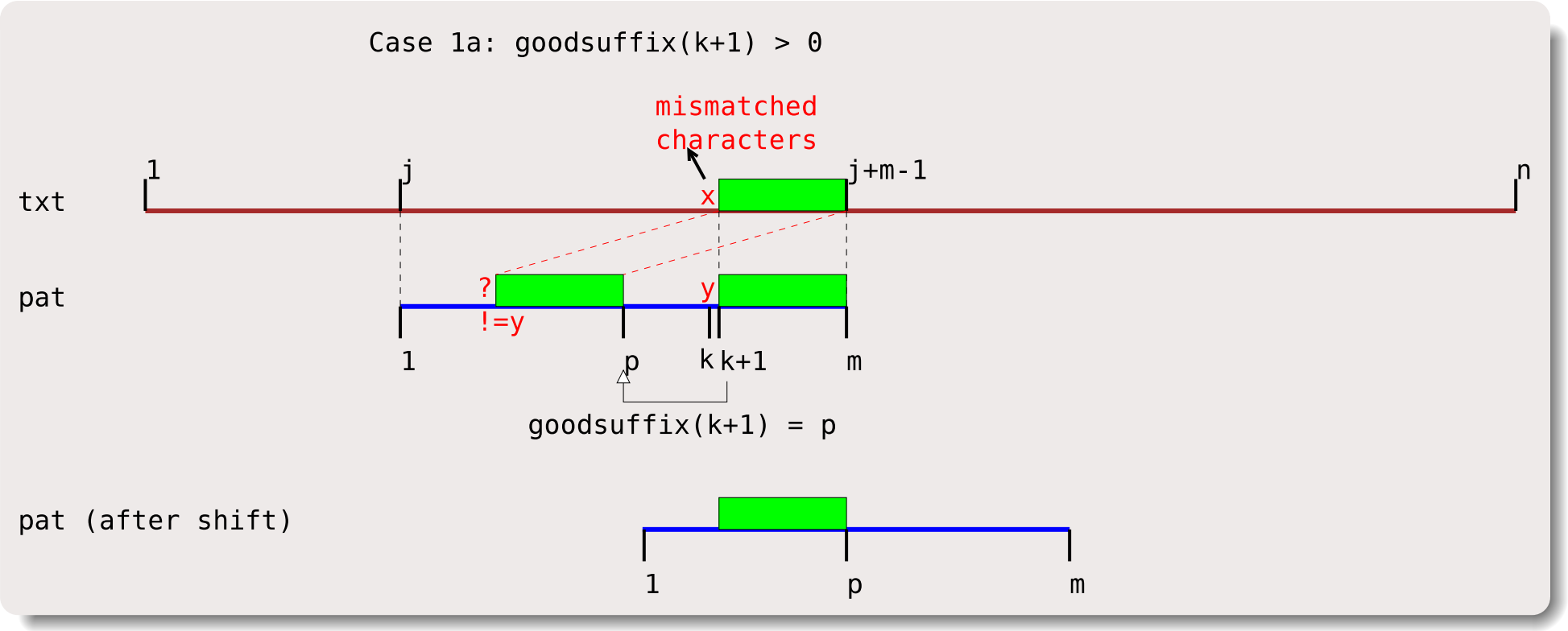
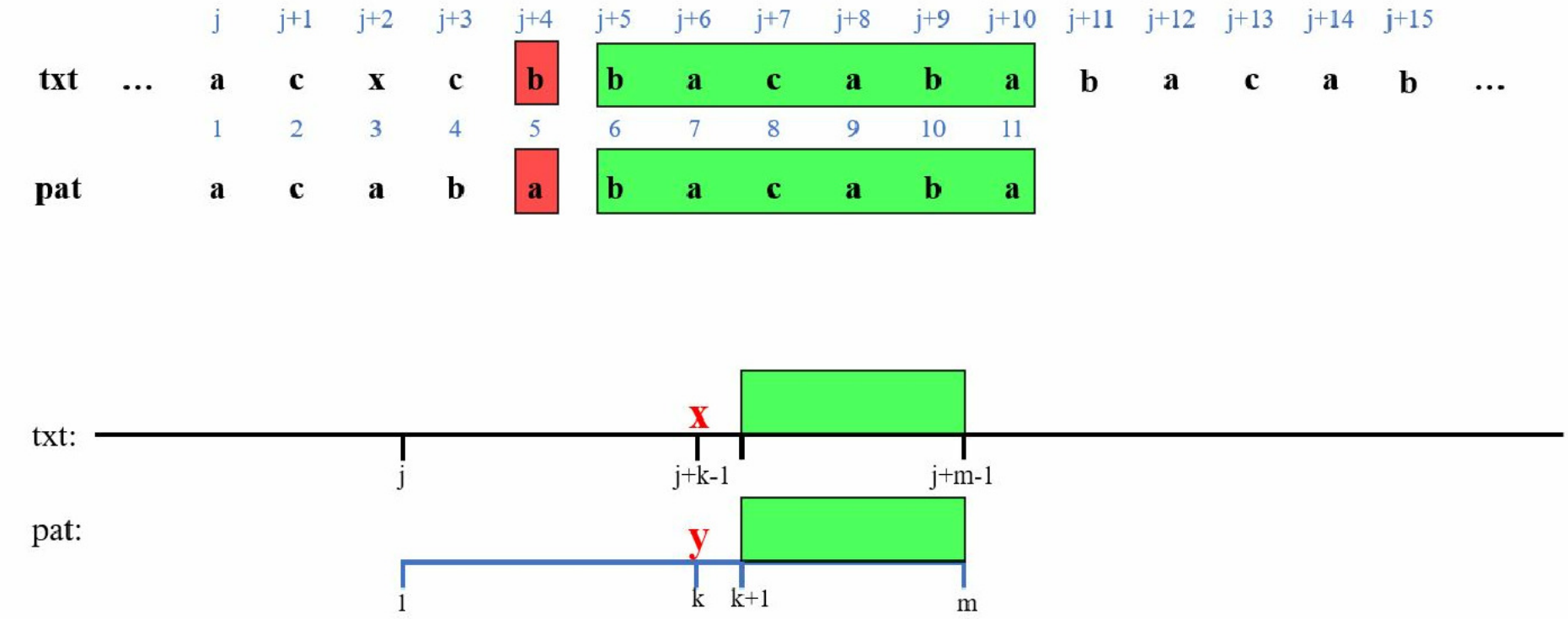


Case 1a: if a mismatch occurs at some **pat**$[k]$, and **goodsuffix**$(k+1) > 0$ then

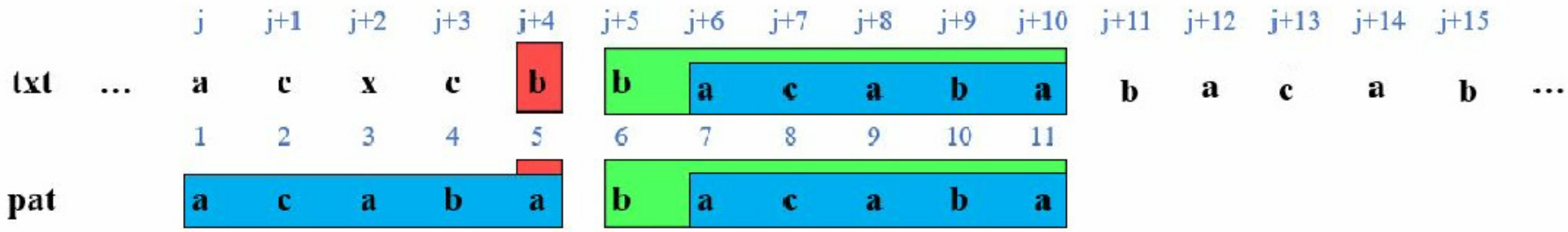
- shift **pat** by $m -$ **goodsuffix**$(k+1)$ places.

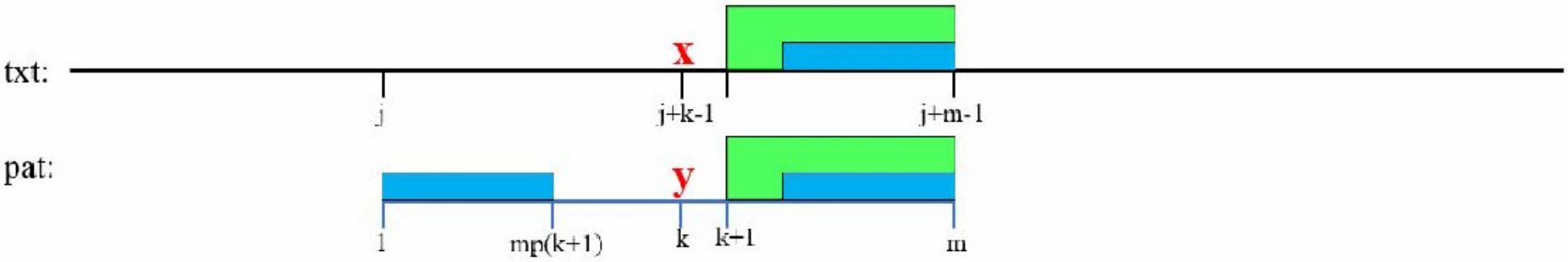
Boyer-Moore Algorithm – Using 'good suffix' rule during search (cont'd)



Case 1b: if a mismatch occurs at some **pat**$[k]$, and **goodsuffix**$(k+1)$=0 then:

Boyer-Moore Algorithm – Using 'good suffix' rule during search (cont'd)

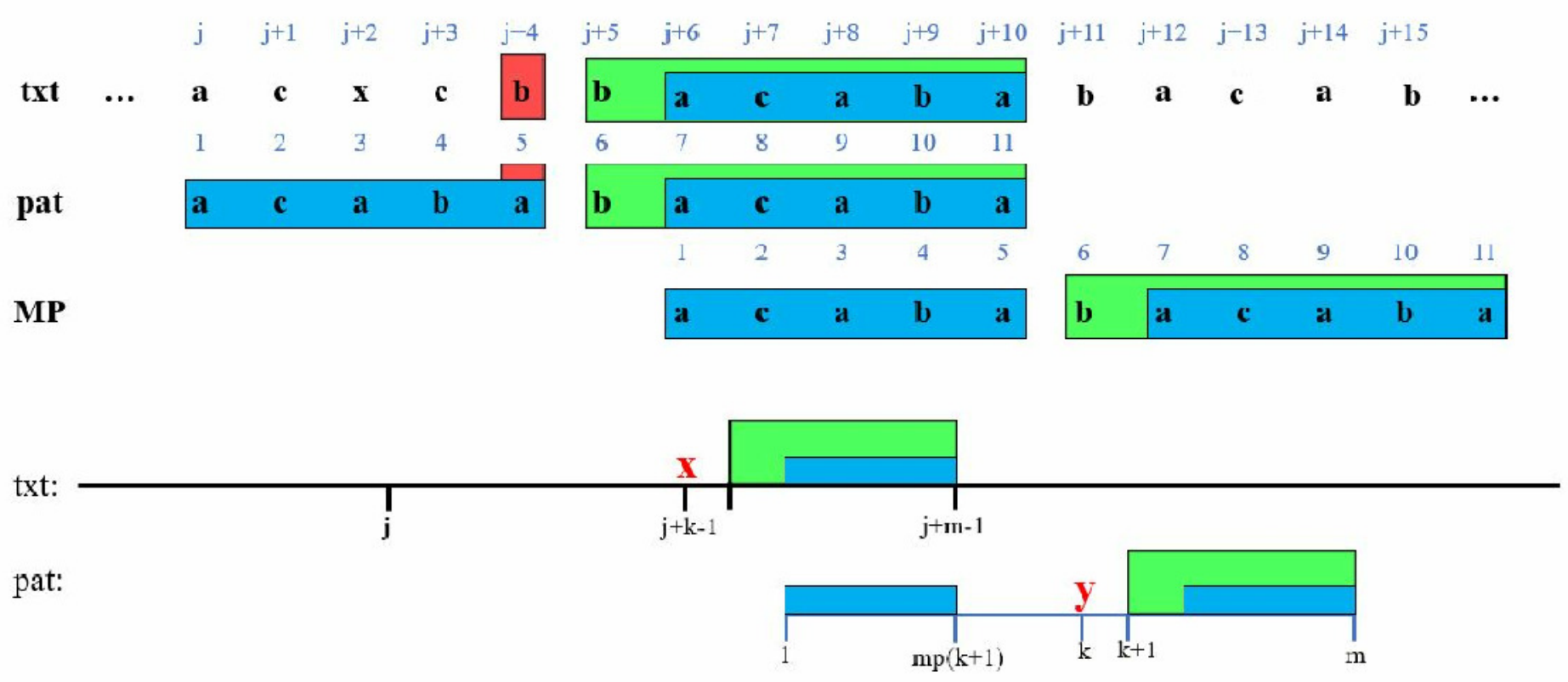




Case 1b: if a mismatch occurs at some **pat**$[k]$, and **goodsuffix**$(k+1)$=0 then:

- **matchedprefix**$(k+1)$ denotes the length of the **largest** suffix of **pat**$[k+1..m]$ that is identical to the prefix of **pat**$[1..m-k]$.
- **matchedprefix**$(.)$ values for **pat** can be precomputed using Z-algorithm in $\mathcal{O}(m)$ time – how?(Hint: z-algorithm – this is a tute question!)

Boyer-Moore Algorithm – Using 'good suffix' rule during search (cont'd)



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Shift **pat** by $m -$ **matchedprefix**$(k+1)$ places

Self-study: Example for **matchedprefix** values

Example of computation of **matchedprefix** values

For the same example as before, **pat**$[1 \ldots 11] =$ a c a b a b a c a b a (positions 1 2 3 4 5 6 7 8 9 10 11), see how the **matchedprefix** values compare to Z^{suffix} and goodsuffix.

j	1	2	3	4	5	6	7	8	9	10	11	12
pat	a	c	a	b	a	b	a	c	a	b	a	
Z^{suffix}	1	0	1	0	5	0	3	0	1	0	-	-
goodsuffix	0	0	0	0	0	0	5	0	7	0	9	10
matchedprefix	11	5	5	5	5	5	5	1	1	1	1	0

Boyer-Moore Algorithm – shift rule when pattern is found

```
txt     . . . a a b a a c a b a c a a b a a c a b a c a a . . .
                | | | | | | | | | | | | | |
pat             a a b a a c a b a c a a b a
                1 2 3 4 5 6 7 8 9 10 11 12 13 14
```

Case 2: when **pat**$[1 \ldots m]$ fully matches **txt**$[j \ldots j + m - 1]$

(i.e., the case when an exact occurrence of **pat** was found in **txt** at a position starting from j)

Boyer-Moore Algorithm – shift rule when pattern is found

```
txt     . . . a a b a a c a b a c a a b a a c a b a c a a . . .

pat                                     a a b a a c a b a c a a b a
                                        1 2 3 4 5 6 7 8 9 10 11 12 13 14
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(i.e., the case when an exact occurrence of **pat** was found in **txt** at a position starting from j)

- shift **pat** by $m -$ **matchedprefix**(2) places. Why? (Reason what **matchedprefix**(2) tells you about the **pat**.)

Boyer-Moore Algorithm – Bringing all pieces together

Preprocessing step

- Preprocess **pat** . . .
  - . . . for the shift data structure (eg. $R(\cdot)$, or $R_k(\cdot)$ values) needed for 'bad-character' shifts (see slides 11-13)
  - . . . for **goodsuffix**$(\cdot)$ and **matchedprefix**$(\cdot)$ values needed for 'good suffix' shifts (see slides 14-22)

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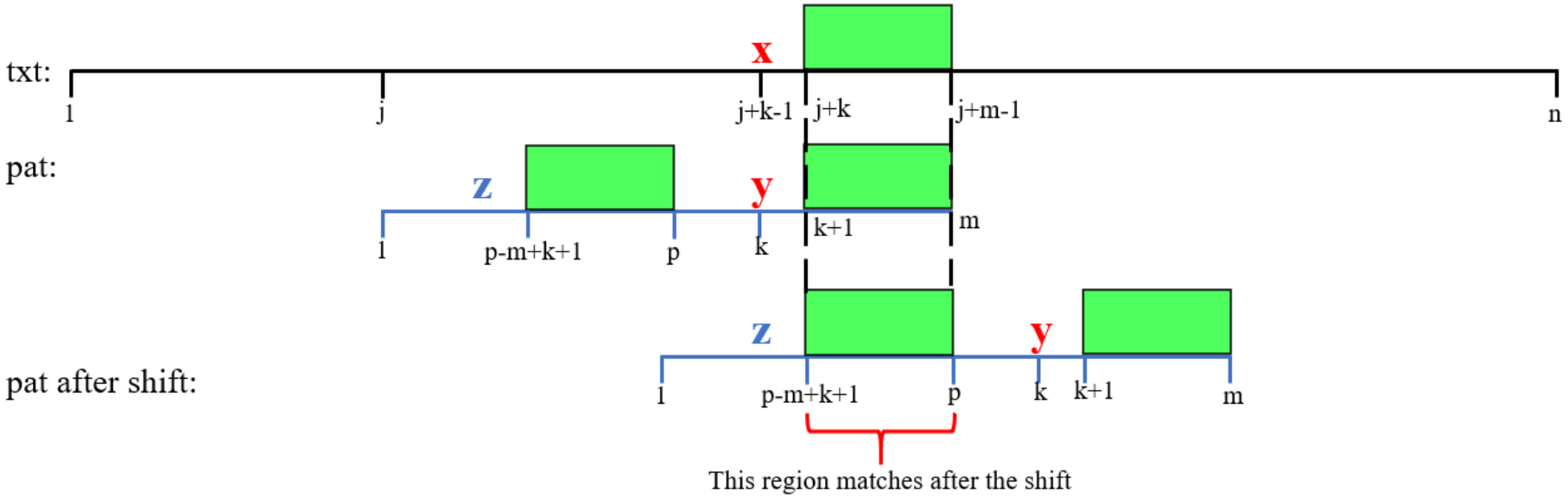
Algorithm

- Starting with **pat**$[1..m]$vs.**txt**$[1..m]$, in each iteration, scan 'right-to-left'
- If, in any iteration, the region matches fully (i.e., **pat** found), shift using case 2 (slide 22)).
- Otherwise (if pattern not found, we have a bad-character and good-suffix):
  - Use (extended) bad-character rule to find how many places to the right **pat** should be shifted under **txt**. Call this shift amount, $s_{\text{badcharacter}}$.
  - Use good suffix rule to find how many places to the right **pat** should be shifted under **txt**. Call this shift amount, $s_{\text{goodsuffix}}$.
  - Shift **pat** to the right under **txt** by $\max(s_{\text{badcharacter}}, s_{\text{goodsuffix}})$ places.

An obvious **optimization** between iterations

Remember!

Between two successive iterations, never recompare character-pairs between **pat** and **txt** that you already know will match. (This certainty forms the very basis of various shifts in this algorithm).



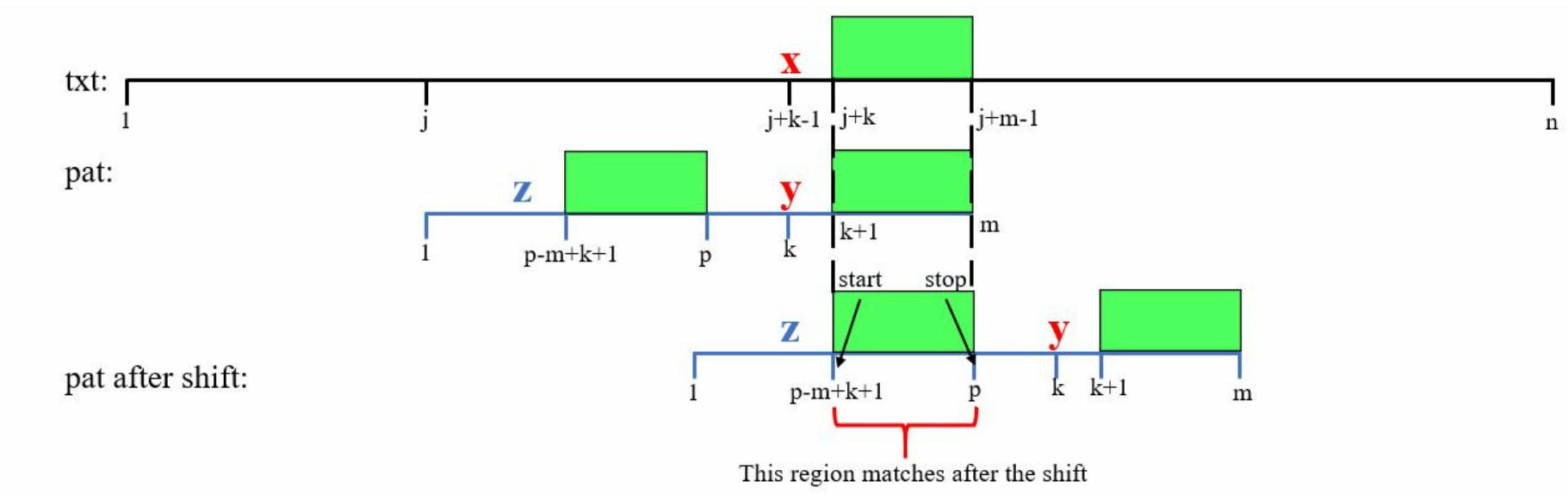
This optimization ensures that **Boyer Moore algorithm** can identify all exact matches of **pat** in **txt** in *worst-case* of $\mathcal{O}(n)$ time, after $\mathcal{O}(m)$ time spent in preprocessing **pat**.

(Non-examinable discussion on time-complexity during the lecture.)

Boyer-Moore Algorithm – Ideas to implement this optimisation

All we need are two variables (say) $start$ and $stop$ to keep track of regions between iterations that are guaranteed to match in the next iteration. For example:

- If **goodsuffix**$(k+1) > 0$, in the lead up to the next iteration, earmark $start =$ **goodsuffix**$(k+1) - m + k + 1$ and $stop =$ **goodsuffix**$(k+1)$.



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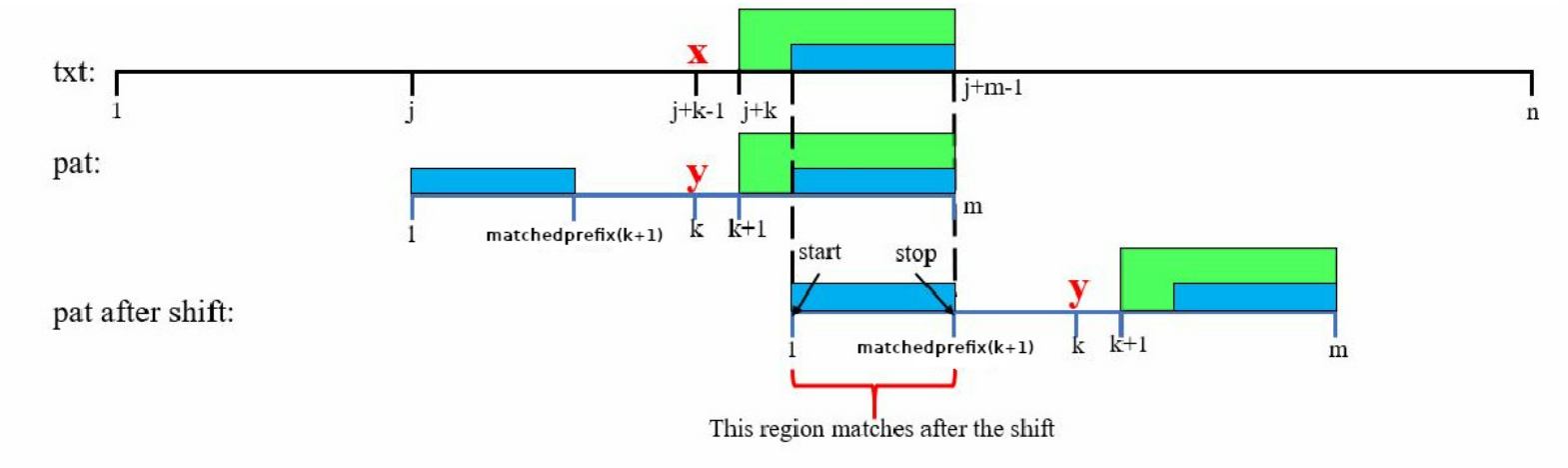
- Else, earmark $start = 1$ and $stop =$ **matchedprefix**$(k+1)$.

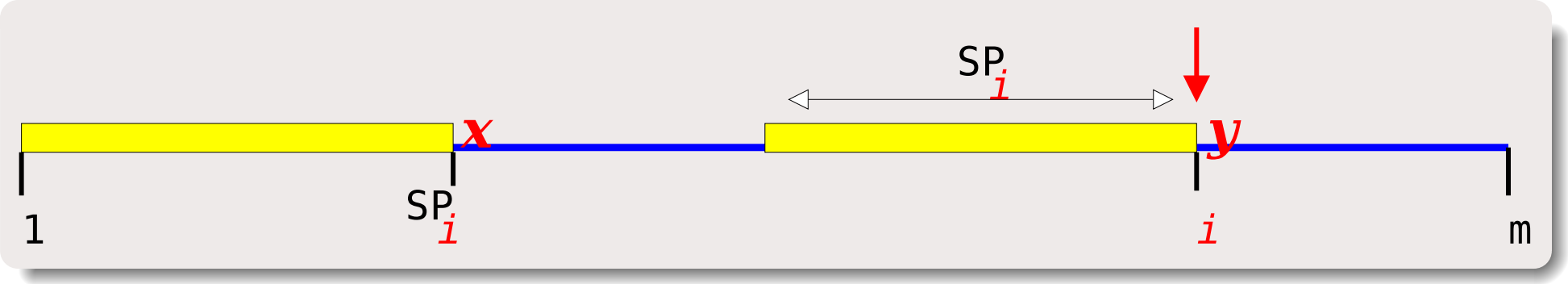


Knuth-Morris-Pratt (KMP) Algorithm

Comments:

- KMP is a very popular algorithm covered in undergrad CS courses.
- Although, it is rarely the method of choice..
- ...and is inferior in performance to Boyer-Moore we discussed above.
- It's explanation is rather simple, as we will see below.
- In fact, much like we did in BM, we will also approach KMP employing the Z-algorithm, which makes the whole implementation even cleaner/better.

KMP algorithm – Defining $\mathbf{SP}_i$ values for $\mathbf{pat}[1 \ldots m]$



Definition of $\mathbf{SP}_i$:

Given a pattern $\mathbf{pat}[1 \ldots m]$, define $\mathbf{SP}_i$ (for each position i in $\mathbf{pat}$) as the **length** of the **longest proper suffix** of $\mathbf{pat}[1 \ldots i]$ that **matches a prefix** of $\mathbf{pat}$, such that $\mathbf{pat}[i+1] \neq \mathbf{pat}[\mathbf{SP}_i{+}1]$.

Example

	1	2	3	4	5	6	7	8	9	10	11	12
pat=	b	b	c	c	a	e	b	b	c	a	b	d

$\mathbf{SP}_1 = 0$ $\mathbf{SP}_2 = 1$ $\mathbf{SP}_3 = 0$ $\mathbf{SP}_4 = 0$ $\mathbf{SP}_5 = 0$ $\mathbf{SP}_6 = 0$

$\mathbf{SP}_7 = 0$ $\mathbf{SP}_8 = 1$ $\mathbf{SP}_9 = 3$ $\mathbf{SP}_{10} = 0$ $\mathbf{SP}_{11} = 1$ $\mathbf{SP}_{12} = 0$

Pseudocode to compute $\mathbf{SP}_i$ values for $\mathbf{pat}[1 \ldots m]$ using Z-algorithm (self-study)

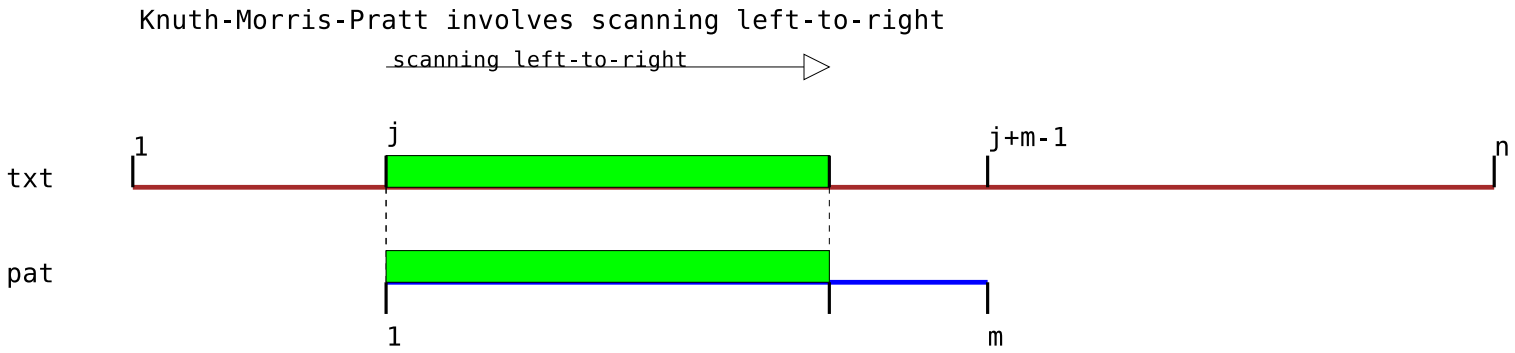
```
Z(1...m) = computeZvalues(pat)
m := |pat|
for i from 1 to m  do
    SP_i := 0
endfor

for j from m down to 2  do
    i := j + Z(j) - 1
    SP_i := Z(j)
endfor
```

KMP Algorithm

- Preprocess the whole **pat** and identify all $\mathbf{SP}_i$ values.
- The general procedure/iteration of KMP involves:

KMP Algorithm



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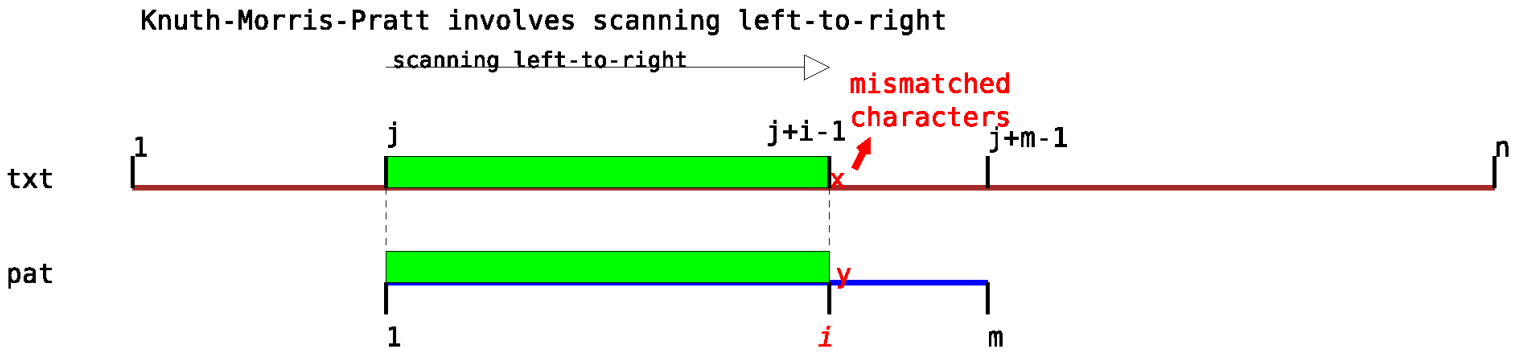
KMP Algorithm



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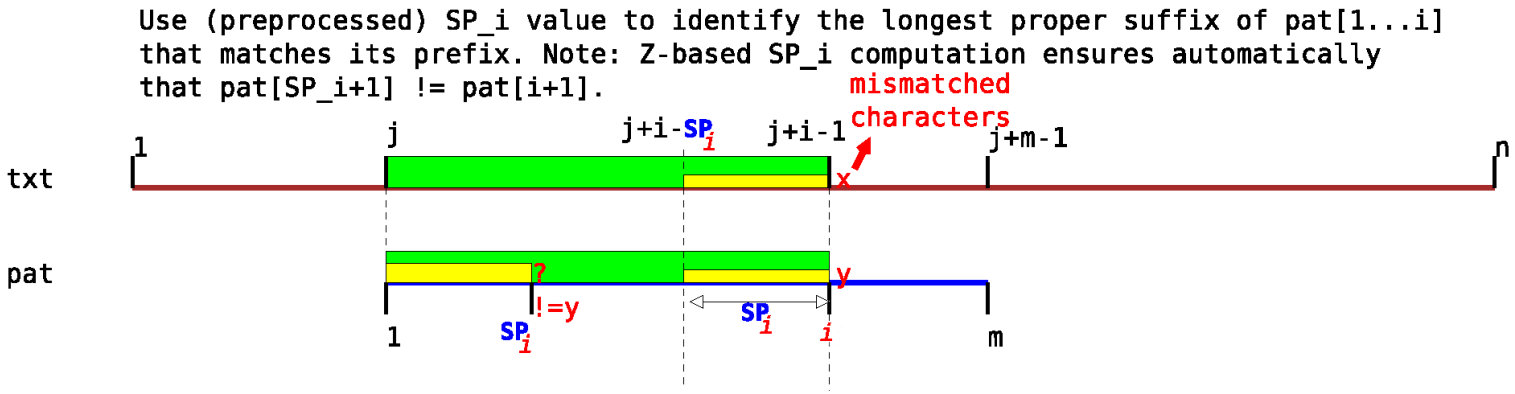

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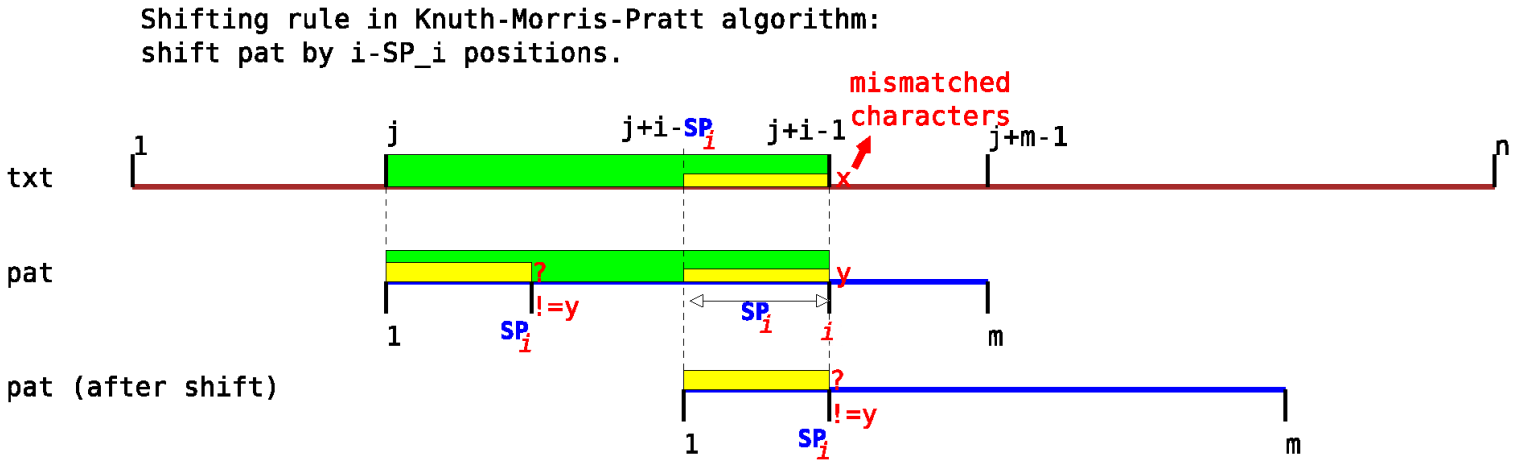
KMP Algorithm



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    - ★ **KMP shift rule** Shift **pat** rightwards by exactly $i - \mathbf{SP}_i$ places so that **pat**$[1 \ldots \mathbf{SP}_i]$ is now aligned with **txt**$[j+i-\mathbf{SP}_i \ldots j+i-1]$.

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  - In the case where an occurrence of **pat** is found in **txt** (i.e., no mismatch), then shift **pat** by $m - \mathbf{SP}_m$ places.

Weeks 1 and 2 summary

- Naïve algorithm takes $\mathcal{O}(mn)$-time, worst case.
- Gusfield's Z algorithm takes $\mathcal{O}(m+n)$-time, worst case.
- Boyer-Moore's algorithm takes
  - $\mathcal{O}(m+n)$-time worst case ...
  - ... but $\mathcal{O}(\frac{n}{m})$-time in most 'realworld' usage.
- Knuth-Morris-Pratt algorithm also takes $\mathcal{O}(m+n)$-time worst-case, but inferior in performance to Boyer-Moore in practice.

Next topic ...

Burrows-Wheeler transform and pattern matching in time proportional to query length.

Preclass learning for Week 3: Revise prefix-doubling suffix array construction learnt in FIT2004.

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END

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